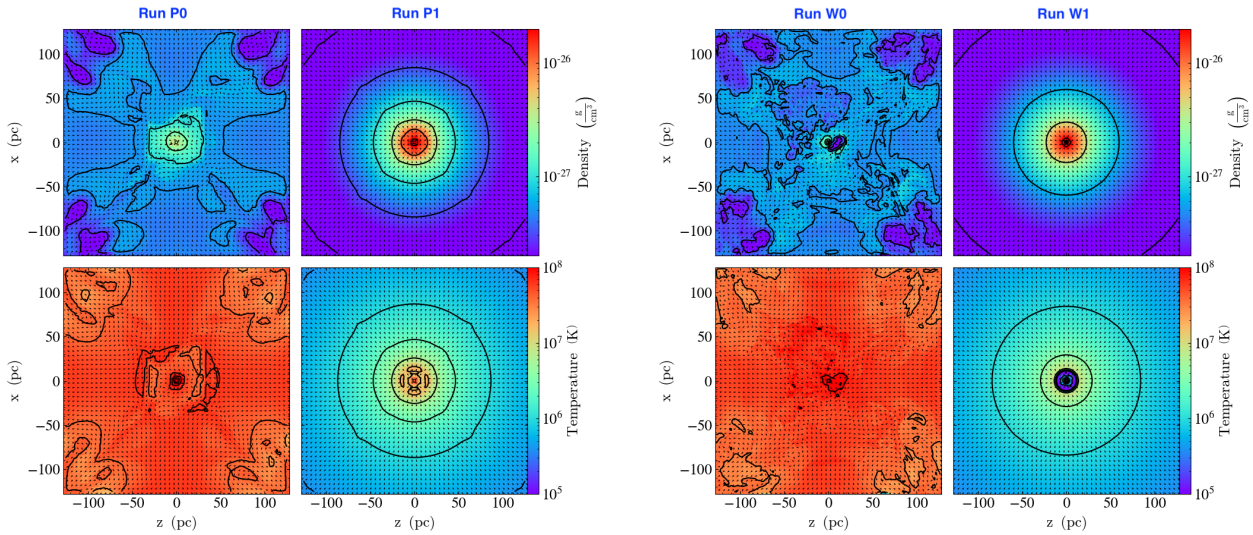

Figure 2. Left: Density and temperature slice plots at $t = 1$ Myr for runs without reset (left) and with kernel reset (right) in the poorly-resolved case (P runs). Right: Density and temperature slice plots at $t = 1$ Myr for runs without reset (left) and with kernel reset (right) for the well-resolved case (W runs).

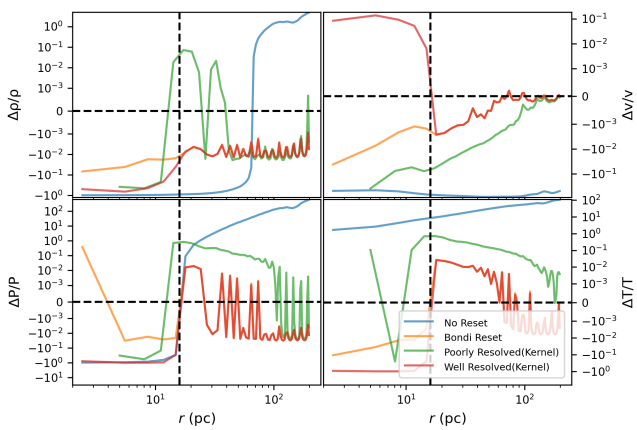

Figure 3. Density, pressure, temperature, and velocity residual profiles of the Bondi test runs, with and without resetting, in comparison with the Bondi solution, taken at $t = 1$ Myr, around 0.9 of the free fall timescale. Plotted here are runs W0 (blue solid line), P1 (green solid line), W1B (orange solid line) and W1 (red solid line). The temperature calculation assumes a 75% hydrogen/25% helium composition. The over-plotted vertical line is the Bondi radius, and the horizontal line represents the ambient value.

radius and the reset radius (both equaling the Bondi radius).

In comparison with a similar published test result (Beckmann et al. 2018), our method is able to better reproduce the accretion rate; we find $|\dot{M}/\dot{M}_{\rm B} - 1| < 0.01$ for the well-resolved setup ($r_{\rm B}/\Delta x = 16$) as compared to their result of $|\dot{M}/\dot{M}_{\rm B} - 1| \sim 0.22$ for the same resolution. We do observe a similar convergence behavior where the gas properties outside the Bondi radius vary with resolution: under-resolved runs tend to underestimate the flow velocity and overestimate pressure and temperature in the few zones around the Bondi radius.

2.2.2. *Bondi-Hoyle-Lyttleton (BHL) Test*

The Bondi problem represents the low Mach number limit of the problem of accretion onto a point mass from a moving uniform medium. In the high Mach number limit, the accretor moves at a speed $v_\infty \gg c_{s,\infty}$ with respect to the medium, and the accretion rate becomes (Hoyle & Lyttleton 1939)

$$\dot{M}_{\rm HL} = \pi r_{\rm HL}^2 \rho_\infty v_\infty \ , \tag{18}$$

where

$$r_{\rm HL} = 2\frac{GM}{v_\infty^2} \ . \tag{19}$$

For the general case an interpolation formula must be used for the accretion rate; one that is accurate to within 10% or so (see Ruffert & Arnett 1994, and references therein) is

$$\dot{M}_{\rm BHL} = 4\pi r_{\rm BHL}^2 \rho_\infty (\lambda^2 c_{s,\infty}^2 + v_\infty^2)^{1/2} \ , \tag{20}$$

where

$$r_{\rm BHL} = \frac{GM}{c_{s,\infty}^2 + v_\infty^2} \ . \tag{21}$$

This is named as the Bondi-Hoyle-Lyttleton problem, and for this test, we use the above result to test the performance of our subgrid model for moving flows. The setup is similar to the Bondi test, but we apply a 45°

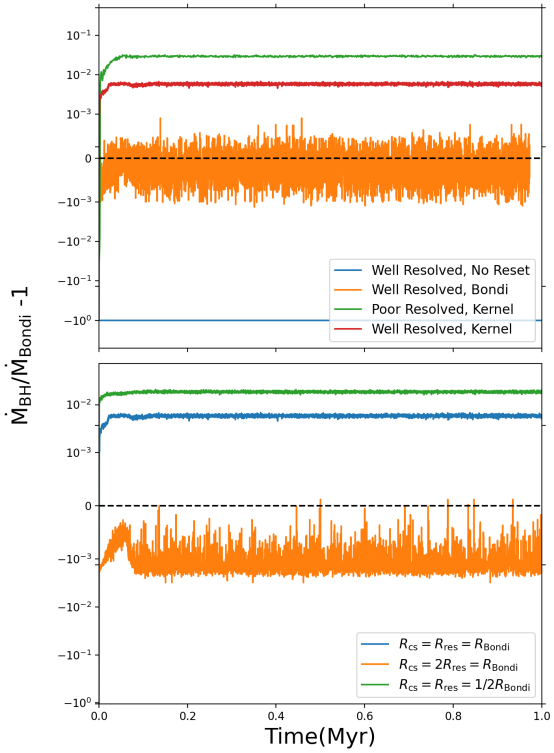

Figure 4. Accretion rate and comparison to the predicted Bondi rate for different Bondi test runs plotted as $\dot{M}/\dot{M}_{\rm B}-1$. This value equals zero when the measured accretion rate is exactly the same as the predicted value. Top panel: Accretion rate with different resetting methods. Plotted here are runs W0 (blue solid line), P1 (green solid line), W1B (orange solid line) and W1 (red solid line). Bottom panel: resetting with a kernel reset for a well-resolved case, but at different accretion and reset radii. Plotted here are W1 (blue), W2 (orange) and W3 (green).

angled inflow boundary condition on two sides of the box and a diode boundary condition on the other sides. While this problem does not have an analytic solution, it has been studied extensively via analytic approximations and numerical simulations; see Edgar (2004) for a review.

To compare with previous work, we ran the test with various black hole masses and Mach numbers $\mathcal{M} \equiv v_\infty/c_\infty$, to demonstrate resolutions for which $r_{\rm BHL}$ is both well-resolved and under-resolved. We also ran tests with different control surface and reset radii, with the full range of run parameters shown in Table 2. However, to ensure enough information is contained on the control surface, we set the minimum control surface and reset radii to be at least two zones across. This means that for runs where the BHL radius is unresolved, resetting is still done, but over a larger region than the gravitational capture radius. The runs are executed on a uniform grid with resolution $\Delta x = 0.25$ pc in a 3D box with size $L_{\rm box} = 128$ pc in each dimension. To encompass a range of resolutions, we set the fiducial black hole mass to be $M = 1.0 \times 10^9\ M_\odot$ and the ambient density to be 1/3 of the value used in the Bondi test, while other initial setup parameters remain the same.

From previous studies, one should expect a bow shock to form in front of the SMBH, raising the temperature of the gas and decreasing its bulk velocity. At radii $r \ll r_{\rm BHL}$, the flow of the gas is approximately radial, and takes the form of a spherically symmetric Bondi solution.

Since the accretion radius changes as the black hole grows, we considered a different approach to setting the control surface radius. This is the "flexible" option in run A1. For the flexible run, we re-evaluated the wind velocity by averaging the gas properties within the Bondi radius of the accretor (which is an upper limit to the BHL radius, and thus large enough to incorporate the flow pattern around the accretor), then updated $R_{\rm cs}$ to correspond to the new values. The new value is further constrained by having the control surface radius remain between $2\Delta x$ and $8\Delta x$, in order to avoid the development of numerical instability. For this problem, the flexible method does not seem to impact the flow measurably compared to run A3. However, in other scenarios where the black hole grows significantly, the flexible option is useful to consider.

For the runs listed in Table 2, we examine the flow pattern and gas dynamics, and we compare the measured accretion rate to Equation (20). In Figures 5, 6, and 7 we show density slices with flow fields perpendicular to the y axis at time $t = 1$ Myr, which is the duration of the run. The flow has mostly reached steady state at this time for all our runs.

Figures 5 and 8 show the gas flow behavior and accretion rate measured through the control surface for the A and B run series (fixing $r_{\rm BHL} = 6.36\Delta x$), varying the control and reset radii. Not surprisingly, varying $R_{\rm cs}$ does not affect the flow structure, since there is no feedback, but the measured accretion rate is different. Using a flexible control surface radius, the run shows that as the steady state solution develops, the accretion radius stays between 2 and $4\Delta x$. The accretion rates are plotted as residuals from the BHL rate (Equation 20). For smaller control surface radii, the measured accretion rate becomes more accurate. For fixed $R_{\rm cs}$, as the reset radius decreases to $2\Delta x$, the accretion rate converges to the BHL value and the shock moves from behind the accretor (unphysical) to in front of the accretor. However, reducing $R_{\rm reset}$ to $1\Delta x$ causes instability to develop behind the accretor that impacts the flow, also making

Table 2. Table of runs conducted for the Bondi-Hoyle-Lyttleton test. Each run used the kernel resetting method with the reset radius and control surface radius indicated, scaled to the resolution of the run.

Objective	Run Number	$M/M_\odot$	Mach Number	$r_{\rm BHL}/\Delta x$	$R_{\rm cs}/\Delta x$	$R_{\rm reset}/\Delta x$
Vary control surface	A1	1.0×10^9	3	6.36	Flexible	2
	A2	1.0×10^9	3	6.36	16	2
	A3	1.0×10^9	3	6.36	8	2
	A4	1.0×10^9	3	6.36	4	2
	A5	1.0×10^9	3	6.36	2	2
Vary reset radius	B1	1.0×10^9	3	6.36	8	8
	B2	1.0×10^9	3	6.36	8	4
	B3 (Same as A3)	1.0×10^9	3	6.36	8	2
	B4	1.0×10^9	3	6.36	8	1
Vary accretion radius	C1	1.0×10^7	3	0.06	2	2
	C2	1.0×10^8	3	0.64	2	2
	C3	3.0×10^8	3	1.91	2	2
	C4 (Same as A5)	1.0×10^9	3	6.36	2	2
	C5	3.0×10^9	3	19.07	2	2
	C6	1.0×10^{10}	3	63.58	2	2
Vary Mach number	D1	1.0×10^9	1	31.79	2	2
	D2 (Same as A5)	1.0×10^9	3	6.36	2	2
	D3	1.0×10^9	5	2.45	2	2
	D4	1.0×10^9	10	0.63	2	2

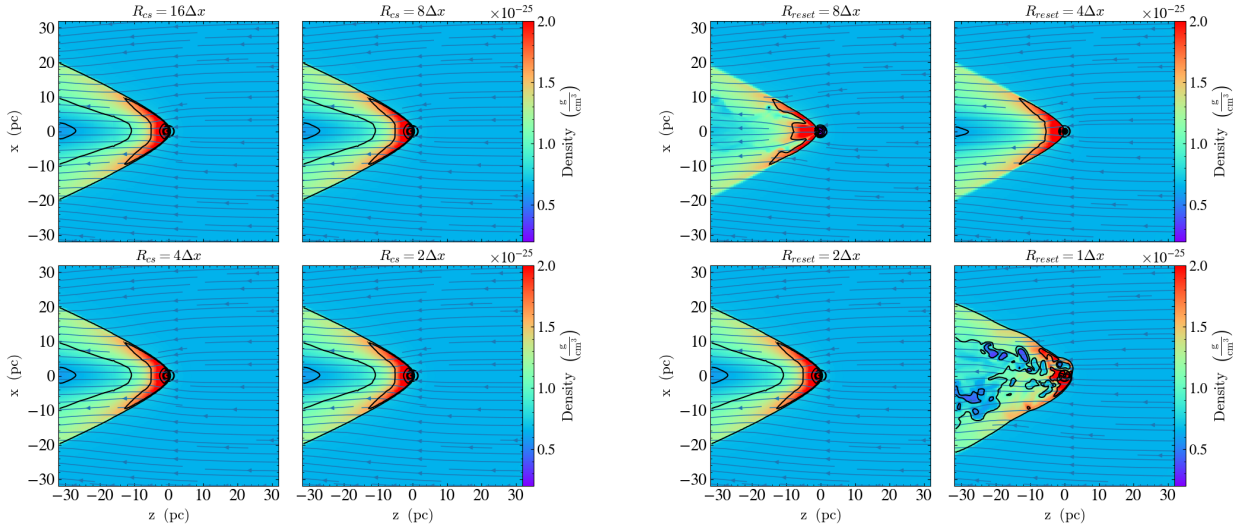

Figure 5. Density slices of the runs with $M = 10^9 M_\odot$ and $\mathcal{M} = 3$ at $t = 1$ Myr (run A1 to B4), plotted within a region of size 64 pc. The left panel shows the A series, and the right panel shows the B series. Overplotted are density contours and velocity streamlines. Black circles represent the accretion radius $r_{\rm BHL}$.